

A link is connectivity between two devices which includes the cables and protocols used in order to make communication between devices.

31) How many layers are in OSI reference model?

OSI reference model: OSI reference model is an ISO standard which defines a networking framework for implementing the protocols in seven layers. These seven layers can be grouped into three categories:

- Network layer: Layer 1, Layer 2 and layer 3 are the network layers.
- Transport layer: Layer 4 is a transport layer.
- Application layer. Layer 5, Layer 6 and Layer 7 are the application layers.

There are 7 layers in the OSI reference model.

1. Physical Layer

- It is the lowest layer of the OSI reference model.
- It is used for the transmission of an unstructured raw bit stream over a physical medium.
- Physical layer transmits the data either in the form of electrical/optical or mechanical form.
- The physical layer is mainly used for the physical connection between the devices, and such physical connection can be made by using twisted-pair cable, fibre-optic or wireless transmission media.

2. DataLink Layer

- It is used for transferring the data from one node to another node.
 - It receives the data from the network layer and converts the data into data frames and then attach the physical address to these frames which are sent to the physical layer.
 - It enables the error-free transfer of data from one node to another node.
- Functions of Data-link layer: